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隱形帳簿：衡量印尼平台交易與收入差距

The Invisible Ledger:

Measuring the Transaction–Revenue Gap of Digital Platforms in Indonesia

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Abstract

This paper examines how public disclosures can support an Indonesia-focused comparison of platform transaction value and revenue. Three FY2023 cases—Grab, Tokopedia e-commerce, and Shopee—combine country disclosures, segment data, and explicitly identified estimates. The baseline constructions give US$43.233 billion of transaction value and US$3.162 billion of selected revenue, a difference of US$40.070 billion. The ecosystem ratio is a transformation of the monetization rate, not an independent measure of undeclared income. Within-source comparisons show that revenue definitions materially affect the results: Tokopedia’s ratio falls from 39.30 to 26.68 when gross rather than net segment revenue is used. Separate historical records document variation in monetization and reporting scope. The findings support transparent measurement of accounting boundaries but do not identify a national tax gap, missing value added, or an investor-response effect.

Keywords: digital platforms; Indonesia; revenue recognition; transaction value; measurement; information reporting.

中文摘要

本研究探討如何利用公開揭露資料，比較印尼數位平台的交易總值與平台收入。主分析限於 2023 年 Grab、Tokopedia 電商及 Shopee 三個案例，並區分國別直接揭露、業務部門資料與推估值。依基準假設，三個案例的交易總值合計為 432.33 億美元，所選收入為 31.62 億美元，差額為 400.70 億美元。生態系比率是收入對交易總值比率的數學轉換，並非未申報所得的獨立衡量。同一來源的比較顯示，收入定義會明顯影響結果：Tokopedia 採用部門總收入而非淨收入時，比率由 39.30 降至 26.68。另列的歷史資料呈現變現程度與揭露範圍的變化。研究結果支持明確揭露衡量方法，但不構成印尼全國稅收缺口、未計入附加價值或投資人反應的估計。

關鍵詞：數位平台、印尼、收入認列、交易總值、衡量、資訊申報。

1. Introduction

Digital platforms coordinate transactions whose value can be much larger than the revenue recognized by the intermediary. A merchant’s sale, a delivery payment, and the platform’s commission describe different economic quantities even when they arise from the same transaction. Multisided-platform research explains why this separation is part of the business model rather than, by itself, evidence of concealment (Rochet and Tirole, 2003; Hagiu and Wright, 2015). The measurement difficulty is deciding which transaction and revenue figures can legitimately be compared.

This paper examines that difficulty using Grab, Tokopedia e-commerce, and Shopee in the Indonesian market. The research question is how public disclosures can be used to construct an Indonesia-focused comparison of transaction value and platform revenue, and how the result changes with assumptions about geography and revenue recognition. The analysis does not begin by assuming that the difference is undeclared income. Instead, it asks what the disclosed quantities establish and where additional assumptions enter.

The Indonesian setting offers three distinct disclosure routes. Grab discloses revenue attributed to Indonesia, but not an Indonesia transaction-value figure. GoTo’s annual report provides transaction value and revenue for the Tokopedia e-commerce segment, with institutional evidence linking that business to Indonesia. Sea discloses Shopee’s company-wide operating and financial information, while Indonesia-specific market size and market share must be obtained from an external estimate. These routes produce different evidentiary strengths; treating all three as directly observed country data would conceal the central measurement problem (Grab, 2024a, 2024b; GoTo, 2024; Sea, 2024a; Santika, 2025).

The main comparison is restricted to fiscal year 2023. Tokopedia’s e-commerce segment, rather than GoTo Group totals, defines the second case; company-wide historical observations remain outside the main Indonesia results. Fiscal year 2023 provides a common period preceding Tokopedia’s 2024 deconsolidation. This is a selected case comparison, not a probability sample of Indonesian enterprises. Its inclusion rule is a documented Indonesia connection and an explicit route to both transaction value and the chosen revenue measure. Admission of a case does not imply that its country allocation or accounting definition has become directly observed.

Under the baseline constructions, the three cases produce US$43.233 billion of transaction value and US$3.162 billion of selected revenue, giving an arithmetic difference of US$40.070 billion. The ratio of the difference to revenue is 12.67. This sum is conditional on the Grab and Shopee assumptions, on treating the Tokopedia segment as Indonesia-aligned, and on combining heterogeneous operating measures. It is not a deduplicated national aggregate, a lower bound on tax non-compliance, or an estimate of Indonesia's unmeasured value added.

The analysis develops three findings. First, country reconstruction and independent country measurement must be distinguished: Grab's and Shopee's ratios are mechanically implied by the revenue-to-transaction rates used to construct a missing input. Second, the magnitude of a ratio depends materially on its accounting boundary. For the same FY2023 Tokopedia transaction value, using gross rather than net third-party segment revenue changes the ratio from 39.30 to 26.68. Third, historical disclosures show substantial variation in monetization and reporting scope, not a single stable ratio that can be transferred across platforms, years, and countries without qualification. These findings concern measurement and comparability; they do not identify an effect on tax reporting or investors.

The fiscal motivation remains relevant but separate. Research on third-party information shows why transaction records can help verify income declarations (Kleven et al., 2011; Pomeranz, 2015; Slemrod, 2019). That literature motivates examining platform records. It does not establish that a residual calculated from public financial statements is missing from a tax authority's information set. The expression "Invisible Ledger" therefore names the motivating information question, not a measured stock of hidden income.

The paper proceeds from the literature and institutional setting to definitions, data construction, and the FY2023 results. Section 6 examines accounting scope and separately reported historical evidence. Section 7 discusses the information-reporting interpretation and the requirements for an investor extension. Source definitions and excluded or unresolved observations are documented in the appendices.

2. Literature and institutional setting

2.1 Platform intermediation and recognized revenue

Rochet and Tirole (2003) and Armstrong (2006) analyze pricing when a platform connects distinct user groups. Participation on one side affects the value received on the other, so prices and commissions need not resemble a conventional markup applied to a vertically integrated producer's output. Hagiu and Wright (2015) distinguish multisided platforms by the control and interaction retained by affiliated participants. These studies provide a rationale for separating the scale of activity coordinated through a platform from the revenue of the platform itself.

They do not, however, supply an empirical tax-gap proxy equal to transaction value minus revenue. Nor does platform theory imply that all companies recognize revenue on the same basis. A marketplace commission, a logistics charge, advertising revenue, financial-service income, and a sale of goods may enter company accounts differently. Grab's FY2023 filing expressly discusses a change from agent to principal accounting for certain delivery offerings. Sea separately reports e-commerce service revenue and total e-commerce segment revenue. These differences matter before an economic interpretation is attached to their ratios (Grab, 2024a; Sea, 2024a).

2.2 Information reporting, informality, and finance

Kleven et al. (2011) show that compliance differs sharply between income covered by third-party information and self-reported income in a Danish audit experiment. Pomeranz (2015) identifies the enforcement role of transaction paper trails in Chile's value-added tax system. Slemrod (2019) distinguishes information reporting, audits, and remittance arrangements as different enforcement instruments. The common implication for the present study is that information about transactions can matter even when the intermediary is not the person liable for tax on the underlying activity.

That implication must not be confused with a measurement result. Public platform accounts do not reveal which sellers have filed returns, what deductible costs they incurred, whether they are below a filing threshold, or which taxes have already been remitted. The informality literature also cautions against treating all nonstandard work or small business activity as one category. La Porta and Shleifer (2014) and Ulyssea (2018) examine dimensions of informality that cannot be recovered simply by subtracting one company aggregate from another.

The question also connects to financial research on platform-enabled work and entrepreneurship. Barrios, Hochberg, and Yi (2022) relate the arrival of gig-work opportunities to new business formation and lending to newly registered firms. Denes, Lagaras, and Tsoutsoura (2025) use U.S. tax records to study transitions from gig work into entrepreneurship. These studies show why platform activity can be financially consequential for participants. They do not validate the ecosystem ratio, establish Indonesian non-compliance, or show that investors price the accounting residual studied here.

The contribution of the present paper is accordingly an application and examination of measurement choices. It reconstructs a limited Indonesian case set, records where public disclosure ends and assumptions begin, and demonstrates the sensitivity of its results to those choices. The arithmetic is not claimed as a new accounting identity or a previously validated tax-compliance measure.

2.3 The three platforms and the selection rationale

Grab combines mobility, deliveries, and financial services. Its FY2023 disclosures attribute US$605 million of revenue to Indonesia. The contemporaneous results release reports both Group gross merchandise value and the narrower sum of Mobility and Deliveries gross merchandise value. The Group figure is used in the retained baseline because the available Indonesia revenue is itself a Group-revenue country allocation; the narrower operating series is used only in a separately labelled company-wide comparison (Grab, 2024a, 2024b).

Tokopedia is the e-commerce business reported within GoTo in FY2023. GoTo's annual report describes it as operating an Indonesian marketplace for physical and digital goods and describes its coverage of Indonesian districts. The report provides an e-commerce operating transaction value and separate gross revenue, customer incentives, and net revenue in the segment note. This is stronger evidence for an Indonesia-focused case than GoTo Group totals, which combine businesses and jurisdictions. Nevertheless, the numbers are business-segment disclosures, not a table explicitly labelled as an Indonesia geographic pair (GoTo, 2024, printed pp. 72 and 142; financial-statement note 29).

Shopee is Sea's e-commerce business. Sea also operates digital entertainment and financial services, so consolidated Sea revenue must not be paired with Shopee transaction value as though they describe the same business. For the Indonesia construction, I use an external estimate of Indonesian e-commerce market size and Shopee's share, then apply Sea's stated company-wide e-commerce service-revenue-to-GMV rate. Sea's broader e-commerce segment revenue is retained for a separate denominator comparison, not silently substituted into the Indonesian baseline (Sea, 2024a).

These platforms are selected because the supplied sources provide an identifiable Indonesian anchor and a reproducible construction for FY2023. This is not a claim that they are the only firms with public information or that they exhaust the market. Other platforms appear in the external market report, but the present source set does not establish the paired revenue measurements required by this design. Adding their market shares alone would enlarge transaction coverage without solving the missing-revenue problem.

3. Measurement framework

3.1 Transaction value, revenue, and the ecosystem ratio

Let V denote a platform's gross transaction value over a specified period and scope, and let R denote the revenue measure selected for that same comparison. I use gross transaction value (GTV) as the generic term in the text. When a source calls its measure gross merchandise value (GMV), the original label and definition are retained in Appendix A. The common notation does not imply that the definitions are identical across providers.[1]

$$W = V - R; \quad E = (V - R) / R; \quad t = R / V. \quad (1)$$

1 GTV is common notation, not a claim of identical company definitions. Grab and Sea use GMV; GoTo uses GTV. Appendix A records the original definitions and selected revenue denominators.

W is the transaction–revenue difference, E is the ecosystem ratio, and t is the revenue-to-transaction ratio, referred to here as the monetization or take rate. “Take rate” is used as an analytical shorthand; it is not necessarily a contractual commission rate. The selected R includes all revenue attributed to Indonesia for Grab, third-party net segment revenue for Tokopedia, and derived service revenue for Shopee. A comparison must retain these labels rather than calling every denominator “net revenue.”

For positive V and R, the following identities apply:

$$E = 1/t - 1; \quad W/V = 1 - t. \quad (2)$$

The ecosystem ratio is an author-defined presentation of the revenue-to-transaction relationship. It contains no additional mathematical information beyond t. If a missing input is constructed using an assumed t, the resulting E is imposed by that assumption. Reporting both measures can improve interpretation, but does not supply two independent empirical results.

The difference W is also not automatically the amount paid to participants. Such an interpretation would require a reconciliation showing how gross orders, completed payments, refunds, taxes, discounts, incentives, product sales, and service charges relate to the selected revenue. A positive residual between aggregates is not enough. The paper therefore uses “transaction–revenue difference” rather than treating “unbooked GTV” as verified merchant or driver income.

3.2 Country allocation and assumption dependence

When country revenue is observed but transaction value is not, a Group rate can be used to construct a country proxy. If R_{ID} is country revenue and t_G is a Group revenue-to-transaction ratio, the construction is:

$$\hat{V}_{ID} = R_{ID} / t_G; \quad \hat{W}_{ID} = R_{ID}(1/t_G - 1). \quad (3)$$

This requires the country’s effective monetization rate to equal the Group rate. It can fail because the country differs in business mix, commission schedules, incentives, recognition policy, or financial-service exposure. Disclosure of the source inputs establishes reproducibility, not the truth of the equality.

For the Shopee construction, let M_{ID} be externally estimated Indonesian market GTV and s be Shopee’s estimated share. Applying the company-wide service rate t_S gives:

$$\hat{V}_{ID} = M_{ID} \times s; \quad \hat{R}_{ID} = M_{ID} \times s \times t_S; \quad \hat{W}_{ID} = M_{ID} \times s(1 - t_S). \quad (4)$$

Market size and market share determine the scale, while the assumed rate determines E. In particular, E remains 9 when t_S equals 10%, regardless of the chosen market size or share. A sensitivity exercise that changes market share therefore cannot validate the 9-fold ratio; it only changes the amounts and the weights of the cases in a combined display.

3.3 Aggregation and interpretation

For a selected set of cases, the aggregate ratio is the sum of the differences divided by the sum of the selected revenues. Equivalently, it is a revenue-weighted average of case-level ecosystem ratios. The blended rate is total selected revenue divided by total selected transaction value. Neither is the simple unweighted average of the three platform rates.

$$E_A = \Sigma(V_i - R_i)/\Sigma R_i = \Sigma[(R_i/\Sigma R_i)E_i]. \quad (5)$$

The arithmetic can be correct even when an economic aggregate is not identified. Grab’s Group-calibrated operating scope includes financial services, while the two other cases concern e-commerce. The source set provides no transaction-level cross-platform deduplication or reconciliation of payment and marketplace activity. The combined display is therefore a conditional arithmetic summary, not a national total. Removing a platform changes that summary’s coverage as well as its number; a leave-one-out result is a composition check, not a new estimate of the same national quantity.

4. Data and sample construction

4.1 Main sample and source hierarchy

The main sample contains three FY2023 Indonesia-focused cases. Table 1 sets out the inclusion rule, observation type, and remaining limitation for each case. All main result and sensitivity tables refer to these cases and this fiscal year. Company-wide histories, other countries, and stock-market observations enter only the separately identified supporting analysis.

Table 1. Main sample and disclosure basis

FY2023 case	Indonesia connection	Observed inputs	Unobserved component
Grab Indonesia	Country revenue attributed to Indonesia	Indonesia revenue; Group revenue and original total GMV	Country GTV is allocated using the Group rate; business mix is not country-isolated.
Tokopedia e-commerce	Indonesian marketplace described in GoTo’s report	E-commerce GTV; third-party net segment revenue	The business segment is not an explicitly reported geographic pair.
Shopee Indonesia	External estimate of Indonesian market size and share	External market inputs; Sea’s company-wide service rate	Indonesia service revenue is derived; country inputs are not issuer disclosures.

Notes: The analytical unit is a platform-year construction, with all three cases referring to FY2023 and an Indonesian target. Consistency of the target does not imply identical measurement quality. Company-wide histories do not enter this sample. Sources: Grab (2024a, 2024b), GoTo (2024), Sea (2024a), and Santika (2025).

The source hierarchy distinguishes an issuer disclosure from an estimate published outside the issuer. Grab’s Form 20-F and results release, GoTo’s annual report, and Sea’s Form 20-F provide company evidence. The World Bank exchange-rate response supplies the annual conversion used for rupiah amounts. The Shopee market inputs are taken from an archived Katadata Databoks article by Santika (2025), which attributes the estimates to Momentum Works. The archived article—not the full commercial research report—is the source directly inspected for those country values. That distinction is retained in the citations.

Table 2 lists the eleven inputs behind the three baseline cases. The count includes a currency conversion and the company-wide figures used to check Sea's stated rate; it is not a count of eleven independent platform observations. Each input is preserved with its source file and locator. Appendix A records the source definitions, and Appendix B distinguishes archived observations from additional usable measurements.

Table 2. Baseline source inputs

Input	Value	Unit	Source / reported scope
Grab Indonesia revenue	605	US$ million	G1; country
Grab Group revenue	2,359	US$ million	G1; Group
Grab Group GMV	20,983	US$ million	G2; original Group basis
Tokopedia e-commerce GTV	248,836,156	IDR million	T1; e-commerce segment
Tokopedia net segment revenue	6,175,190	IDR million	T1; third-party segment
Annual exchange rate	15,236.884662	IDR / US$	F1; Indonesia, 2023
Indonesia e-commerce market	53.8	US$ billion	M1; external estimate
Shopee Indonesia market share	40.0	Percent	M1; external estimate
Shopee company GMV	78.5	US$ billion	S1; company-wide
Shopee service revenue (rounded)	7.9	US$ billion	S1; company-wide
Shopee stated service rate	10.0	Percent	S1; company-wide

Notes: Source keys and precise locators appear in Appendix A. IDR denotes Indonesian rupiah. Original units are retained in this input audit table; analytical dollar amounts elsewhere use U.S. dollars. The baseline uses the stated 10.0% rate, not 7.9/78.5 calculated from rounded amounts. The eleven inputs construct three cases; they are not eleven independent observations.

The monetary results are displayed in U.S. dollars. For Tokopedia, an amount reported in millions of rupiah is divided by the annual IDR-per-USD rate and by 1,000 to obtain billions of U.S. dollars. Source-currency values remain in the input table because they permit direct reconciliation to the annual report. Ratios are calculated from unrounded inputs before display rounding. No current exchange rate is applied to a historical flow.

4.2 Grab Indonesia

Grab reports FY2023 Indonesia revenue of US$605 million, Group revenue of US$2,359 million, and Group GMV of US$20,983 million. The implied Group rate is 2,359/20,983 = 11.2424%. Dividing country revenue by this rate gives an Indonesia GTV proxy of US$5.381397 billion. The corresponding difference is US$4.776397 billion, and the ecosystem ratio is 7.8949 (Grab, 2024a, 2024b).

This construction is not an observed Indonesia GTV series. It allocates Group transaction value in proportion to country revenue. In algebraic terms, it assigns Indonesia the same fraction of Group GTV as its fraction of Group revenue. Consequently, it cannot independently establish that Indonesia has the same business mix or monetization as the Group. The calculation is retained as a transparent baseline for review, with the country-allocation assumption and scope limitation visible.

The original FY2023 Group GMV includes financial-services activity and differs from the On-Demand measure. Replacing the Group denominator with On-Demand GTV while retaining all Indonesia revenue would introduce a different mismatch. The paper does not do so. Section 6 instead shows a wholly company-wide comparison in which both On-Demand transaction value and the Mobility plus Deliveries revenue denominator are observed together.

4.3 Tokopedia e-commerce

The GoTo annual report’s operating-metrics table reports FY2023 e-commerce GTV of IDR248,836,156 million. Note 29 reports third-party net e-commerce segment revenue of IDR6,175,190 million. At the retained World Bank annual rate of IDR15,236.884662 per U.S. dollar, these become US$16.331170 billion and US$0.405279 billion. Their difference is US$15.925891 billion, giving an ecosystem ratio of 39.2961 (GoTo, 2024; World Bank, n.d.).

The GTV locator is printed annual-report page 142, corresponding to PDF page 144. The net-revenue locator is financial-statement page 5/111, corresponding to PDF page 475. Those are different tables. The institutional description on printed page 72 supports treating Tokopedia as Indonesia-aligned, but does not turn the segment note into a separately disclosed geographic pair. The segment also includes physical and digital goods, which matters when it is compared with an external market estimate.

The denominator is revenue after customer incentives, not gross segment revenue. This choice is retained from the review package. Section 5 reports an alternative gross-revenue calculation from the same note to show how much the ratio depends on that accounting choice. The alternative is a diagnostic, not a correction silently applied to the baseline.

4.4 Shopee Indonesia

The archived country-market article reports FY2023 Indonesian e-commerce GMV of US$53.8 billion and a Shopee share of 40%. Their product is US$21.52 billion. Sea’s FY2023 Form 20-F states that e-commerce service revenue divided by GMV was 10.0%. Applying that rate gives derived Indonesia service revenue of US$2.152 billion and a difference of US$19.368 billion (Santika, 2025; Sea, 2024a).

The 10.0% value is the issuer's stated rounded rate. The baseline does not manufacture extra precision by dividing the rounded US$7.9 billion and US$78.5 billion headline values. The more precise service-revenue line is relevant to checking the rounding convention, but it is not substituted into the existing country construction. This preserves the convention in the dataset prepared for advisor review and makes its rounding explicit.

Two qualifications are essential. The country-market amounts are external estimates, not Sea geographic disclosures. In addition, the revenue proxy concerns e-commerce services; Sea's total e-commerce segment also includes sales of goods. The Shopee residual must therefore be described as a service-revenue-based difference. It is not the amount excluded from all Shopee revenue, and its comparison with a net-revenue denominator for Tokopedia is descriptive rather than a fully harmonized accounting comparison.

5. FY2023 Indonesia results and sensitivity

5.1 Baseline measurements

Table 3 presents the three case calculations. Shopee contributes the largest transaction amount and dollar difference; Tokopedia has the largest ratio relative to the selected revenue. These statements answer different questions. A high ratio can result from a small revenue denominator and need not imply a larger transaction amount, a larger informal sector, or lower tax compliance.

Table 3. FY2023 Indonesia-focused measurements

Case	GTV	Selected R	Difference	E	t (%)
Grab Indonesia	5.381	0.605	4.776	7.89	11.24
Tokopedia e-commerce	16.331	0.405	15.926	39.30	2.48
Shopee Indonesia	21.520	2.152	19.368	9.00	10.00
Selected-case sum	43.233	3.162	40.070	12.67	7.31

Notes: Monetary amounts are US$ billion. Difference = GTV − selected R; E = difference/R; t = R/GTV. Grab uses all Indonesia revenue and derived GTV; Tokopedia uses net third-party segment revenue; Shopee uses derived service revenue. The sum is an undeduplicated, heterogeneous arithmetic summary, not an Indonesia-wide total. Ratios use unrounded inputs; displayed components may differ slightly because of rounding. Sources: Table 2 and author calculations.

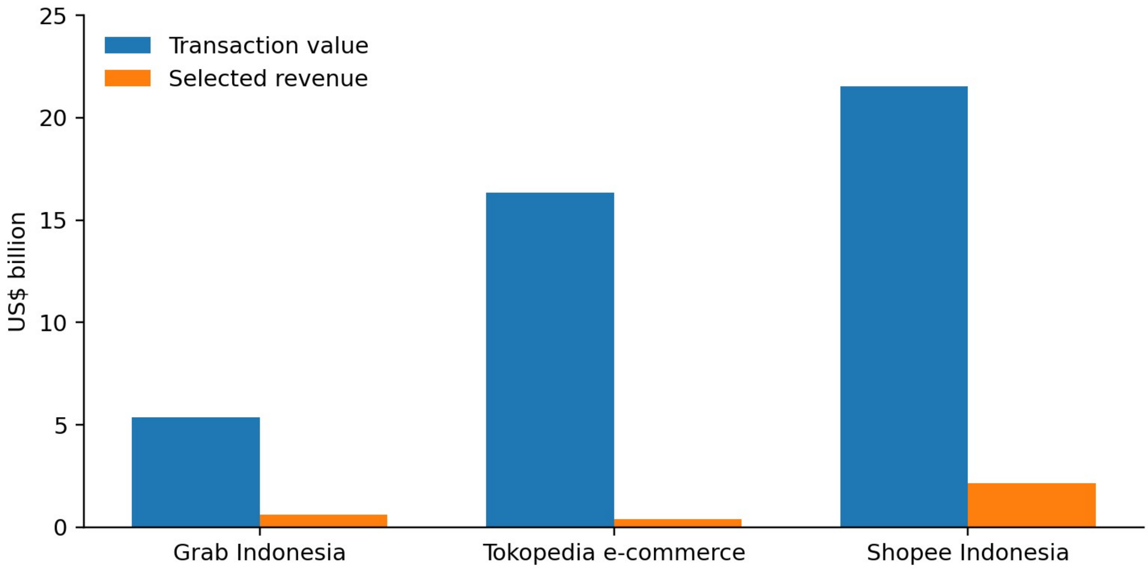

Figure 1. FY2023 platform case measurements

Notes: Transaction value and selected revenue are shown in US$ billion for the three Indonesia-focused constructions. Revenue means all country-attributed revenue for Grab, net third-party segment revenue for Tokopedia, and derived service revenue for Shopee. Source: Table 3. No national total is plotted.

The sum of the selected-case differences is US$40.070 billion and the blended rate is 7.315%. Tokopedia contributes about 37.8% of the selected transaction amount but only 12.8% of selected revenue. Its low net-revenue-to-GTV ratio therefore has a large effect on the aggregate relationship. This is a weighting consequence of the chosen revenue definition and case composition, not evidence that the same ratio characterizes the wider Indonesian economy.

For Grab and Shopee, the baseline ratios should be read as calibration outputs. Grab's 7.89 ratio follows from the Group rate used to derive GTV; Shopee's 9.00 follows from the assumed 10% service rate. Tokopedia's 39.30 is computed from two disclosed segment quantities, although its geographic interpretation and the incentives treatment still require attention. Treating all three ratios as independent estimates of an underlying national parameter would overstate the evidence.

5.2 Assumption sensitivity

Table 4 summarizes the one-way scenarios already supplied with the review package. Each changes one assumption while holding the remaining baseline inputs fixed. The ranges are chosen scenarios, not confidence intervals, estimated error distributions, or evidence that the assumed country rates are correct. The five Grab scenarios are also included among the fourteen one-way scenarios; they should not be counted as nineteen distinct tests.

Table 4. One-way assumption sensitivity

Assumption varied	Scenario range	Rows	Sum of differences	Aggregate E
Grab rate multiplier	0.8–1.2 × Group rate	5	39.173–41.416	12.39–13.10
Shopee market share	35–45%	3	37.649–42.491	12.38–13.01
Shopee service rate	9–11%	3	39.855–40.285	11.80–13.67
Indonesia market size	0.9–1.1 × US$53.8bn	3	38.133–42.007	12.44–12.94

Notes: The baseline sum is US$40.070bn. Dollar ranges are US$ billion; each scenario changes one parameter only. Ranges are scenarios, not confidence intervals. The five Grab-rate scenarios are part of the fourteen rows summarized here, not additional observations. All cases retain the FY2023 Indonesian target. Source: archived one-way sensitivity file and author summaries.

Changing Grab's rate multiplier from 0.8 to 1.2 changes its GTV inversely and moves the selected-case difference from US$41.416 billion to US$39.173 billion. Changing Shopee's share from 35% to 45% moves the difference from US$37.649 billion to US$42.491 billion. In contrast, varying Shopee's service rate from 9% to 11% changes its revenue directly while leaving externally estimated country GTV fixed. These exercises separate sensitivity to transaction scale from sensitivity to retained revenue.

A positive difference survives all the displayed scenarios, but this is not a strong empirical test of hidden activity. If an assumed rate remains below one, equations (3) and (4) already imply a positive difference. The informative part is the change in magnitude and composition, not the survival of the sign. None of the one-way ranges captures all possible misspecification of the transaction definitions, geography, or overlapping activities.

Table 5 removes each platform in turn. The remaining differences range from approximately US$20.70 billion to US$35.29 billion. Excluding Tokopedia reduces the aggregate ratio to about 8.76, illustrating the effect of the low net-revenue denominator. Excluding Shopee raises the remaining ratio because Shopee supplies a large share of revenue at a lower case-level ratio than Tokopedia. These are different coverage sets, not interchangeable country estimates.

Table 5. Leave-one-platform-out comparisons

Excluded case	GTV	Selected R	Difference	E
Grab	37.851	2.557	35.294	13.80
Tokopedia	26.901	2.757	24.144	8.76
Shopee	21.713	1.010	20.702	20.49

Notes: Amounts are US$ billion. Each row contains the other two FY2023 cases. Exclusion changes coverage and is not an alternative estimate of one common national aggregate. Source: archived leave-one-platform-out output; arithmetic reproduced without altering the baseline.

5.3 Tokopedia's gross-versus-net denominator

The same FY2023 segment note allows a more directly observed accounting diagnostic. Tokopedia's third-party gross segment revenue is IDR8,988,909 million, customer incentives are IDR2,813,719 million, and third-party net segment revenue is IDR6,175,190 million. The gross-minus-incentives reconciliation is exact. Holding GTV fixed, changing only the revenue definition reduces the ecosystem ratio from 39.30 to 26.68 (GoTo, 2024, note 29).

Table 6. Tokopedia revenue-basis comparison

Revenue basis	R	Difference	E	t (%)
Net third-party revenue	0.405	15.926	39.30	2.48
Gross third-party revenue	0.590	15.741	26.68	3.61

Notes: The same FY2023 Indonesia-aligned e-commerce GTV of US$16.331bn is used in both rows. Monetary amounts are US$ billion. Gross revenue less customer incentives equals net revenue: IDR8,988,909m − IDR2,813,719m = IDR6,175,190m. The gross-revenue row is an added diagnostic, not a replacement for the main net-revenue convention. Source: GoTo (2024), note 29, PDF p.475, and author calculations.

This is not evidence that one denominator reveals the true hidden economy. It demonstrates that the size of the ratio responds to a disclosed accounting treatment even without changing the platform, year, or transaction measure. A net-revenue residual also contains the arithmetic effect of customer incentives. It cannot automatically be interpreted as merchant receipts. The result strengthens the case for reporting the revenue definition next to the ratio rather than ranking platforms on ratio size alone.

5.4 Scale relative to GDP

Following the request to put the amount in macroeconomic context, I compare it with the official FY2023 nominal gross domestic product (GDP). BPS’s 5 February 2024 release reports GDP at current market prices of IDR20,892.4 trillion. Applying the same annual exchange rate used elsewhere produces approximately US$1,371.173 billion. The selected-case transaction–revenue difference is numerically equivalent to 2.92% of that denominator (BPS-Statistics Indonesia, 2024; World Bank, n.d.).

This is a scale comparison only. GDP measures value added, while the numerator is a difference between selected gross transaction metrics and selected revenue measures. The 2.92% figure does not measure the platforms’ contribution to GDP, an omitted share of GDP, or the size of a recoverable tax base. It is not an additional national estimate, and the lack of transaction-level deduplication remains relevant. The denominator and calculation are included in the supplementary calculation file rather than used as a new headline claim.

6. Accounting scope and separate historical evidence

6.1 Same-year company-wide comparisons

The first five sections keep the main geographic and time target fixed. This section changes the question: it examines what company-wide disclosures reveal about accounting definitions and their evolution. The observations are not added to the Indonesia sample and do not independently validate the country-allocation assumptions.

Grab's FY2023 release reports Group GMV of US$20.983 billion and Group revenue of US$2.359 billion. It separately reports On-Demand GMV of US$15.592 billion, Deliveries revenue of US$1.194 billion, and Mobility revenue of US$0.869 billion. The corresponding ecosystem ratios are 7.89 for the Group pair and 6.56 for the On-Demand pair. Both calculations are company-wide and use the same FY2023 release; the difference is not an observed Indonesia-versus-Group effect (Grab, 2024b).

Sea's FY2023 filing similarly distinguishes a stated service monetization rate of 10.0% from total e-commerce segment revenue of US$9.000848 billion. Relative to US$78.5 billion of GMV, the stated service-rate construction implies 9.00, while the total-segment pair implies 7.72. These are not rival estimates of a single fully harmonized quantity. They describe different revenue boundaries. Table 7 places the company-wide alternatives together so that their geographic scope is unambiguous (Sea, 2024a).

Table 7. Company-wide FY2023 scope diagnostics

Company	Measurement basis	GTV	R	E
Grab	Group total	20.983	2.359	7.89
Grab	Mobility + Deliveries	15.592	2.063	6.56
Shopee	Stated 10% service rate	78.500	7.850*	9.00
Shopee	Total e-commerce segment	78.500	9.001	7.72

Notes: These are company-wide observations, not Indonesia measurements. Monetary amounts are US$ billion. *US$7.850bn is implied by 78.5 × 10%, not the directly reported service-revenue amount. The precise disclosed service revenue is US$7.885185bn, while total e-commerce revenue is US$9.000848bn; the table preserves the baseline's stated-rate convention for comparison. Grab's narrower pair includes Mobility and Deliveries only. Sources: Grab (2024b); Sea (2024a); author calculations.

These diagnostics also explain why later restated annual figures, contemporaneous quarterly figures, and the Indonesia proxy cannot simply replace one another. A "matched" pair in this paper means that the numerator and denominator refer to an explicitly documented business scope and period within one provider. It does not mean that the analysis has converted the three firms to an identical business model or adjusted their accounts to a common recognition policy.

6.2 Earlier annual disclosures

Table 8 presents the eight earlier annual records retained in the source package: Grab Mobility plus Deliveries for 2019–2021 and Shopee e-commerce for 2017–2021. Only seven records yield interpretable positive-denominator ratios. Grab's 2019 matched segment revenue is negative after incentive netting; it remains in the table as an accounting observation, with the ratio marked not meaningful rather than reported as a very large economic quantity (Grab, 2022; Sea, 2019–2022).

Table 8. Earlier company-wide annual records

Company	Year	GTV	R	E
Grab	2019	8.662	-0.629	n.m.
Grab	2020	8.700	0.443	18.64
Grab	2021	11.317	0.604	17.74
Shopee	2017	4.113	0.009	454.27
Shopee	2018	10.279	0.270	37.13
Shopee	2019	17.576	0.834	20.07
Shopee	2020	35.400	2.167	15.33
Shopee	2021	62.600	5.123	11.22

Notes: All observations are company-wide, not Indonesia-specific. Grab uses Mobility plus Deliveries; Shopee uses e-commerce. Amounts are US$ billion; ratios use the underlying unrounded source extracts. The eight records supply seven interpretable ratios; n.m. denotes a non-meaningful ratio because Grab's FY2019 revenue is negative. Sources: Grab (2022); Sea (2019, 2020, 2021, 2022); archived historical extracts.

Shopee's ratio falls from approximately 454.27 in 2017 to 11.22 in 2021 while both transaction value and revenue increase. The earlier ratio reflects a very small revenue denominator. The annual series therefore shows changes in monetization rather than a stable universal multiplier. Grab's 2020 and 2021 ratios are approximately 18.64 and 17.74, but the negative 2019 denominator cautions against treating the entire sequence as a smooth economic trend.

The separate Indonesia extension contains nine archived platform-year records, including the three FY2023 baseline cases. Appendix B displays their coverage without treating those overlaps as new evidence. It also withholds the Tokopedia FY2021 ratio from the active analysis. That row combines full-year pro-forma GTV with IDR1,997,222 million of revenue that the FY2022 annual report identifies as the post-acquisition contribution from 17 May 2021. A common FY2021 label does not make those periods equivalent (GoTo, 2023, business-combination note, PDF p. 494).

The FY2022–FY2023 Tokopedia comparison is more informative because the FY2023 report supplies the comparative GTV and restated net segment revenue together. GTV falls by 8.90%, net segment revenue rises by 53.20%, and the source-currency transaction–revenue difference falls by 9.83%. The ratio falls from 66.76 to 39.30. This is descriptive evidence of changing recorded monetization, not a measure of improved tax compliance. No missing full-year FY2021 net-revenue figure is invented to lengthen the series.

6.3 The quarterly accounting panel

The retained quarterly file contains 47 platform-quarter accounting records: 17 for Grab, 16 for GoTo, and 14 for Sea. Grab and GoTo run to 2026Q1 from 2022Q1 and 2022Q2 respectively. Sea includes four 2022 quarters, 2023Q4, all four quarters in 2024 and 2025, and 2026Q1. Sea's first three 2023 quarters are absent from the retained panel; they are not interpolated. An absent row is not treated as a zero or as proof that no relevant information could ever be recovered.

Table 9 summarizes the retained ratios by year and platform. It reports means of the available quarterly ratios, not ratios of annual totals. The 2023 Sea entry contains only Q4, and the 2026 entries contain only Q1. Each platform's geographic scope is company-wide: Grab's Mobility and Deliveries operations, GoTo Group on the stated release basis, and Sea's Shopee e-commerce operations. These are not three Indonesian country panels.

Table 9. Quarterly ratios by company and year

Company	Year	N	Mean E	Median E	Range
Grab	2022	4	10.61	10.03	6.66–15.73
Grab	2023	4	6.56	6.54	6.39–6.79
Grab	2024	4	6.23	6.23	6.11–6.35
Grab	2025	4	6.32	6.34	6.08–6.54
Grab	2026*	1	6.24	6.24	6.24–6.24
GoTo	2022	3	53.08	46.91	34.22–78.11
GoTo	2023	4	40.22	40.08	37.14–43.58
GoTo	2024†	4	34.38	33.76	33.14–36.85
GoTo	2025	4	36.22	35.29	33.17–41.12
GoTo	2026*	1	43.25	43.25	43.25–43.25
Shopee / Sea	2022	4	9.35	9.61	7.57–10.60
Shopee / Sea	2023*	1	7.88	7.88	7.88–7.88
Shopee / Sea	2024	4	7.16	7.08	6.73–7.74
Shopee / Sea	2025	4	6.71	6.67	6.34–7.17
Shopee / Sea	2026*	1	6.31	6.31	6.31–6.31

Notes: N counts retained quarterly accounting records, not country observations. Means and medians are of quarterly ratios, not ratios of annual totals. *Sea 2023 contains Q4 only; 2026 contains Q1 only for each company. †GoTo 2024 is on the release's comparable/pro-forma basis. Grab is Mobility plus Deliveries, Sea is Shopee e-commerce, and GoTo is Group-level. Source: retained 47-row accounting panel and issuer-source reconciliation; no stock-return inference is reported.

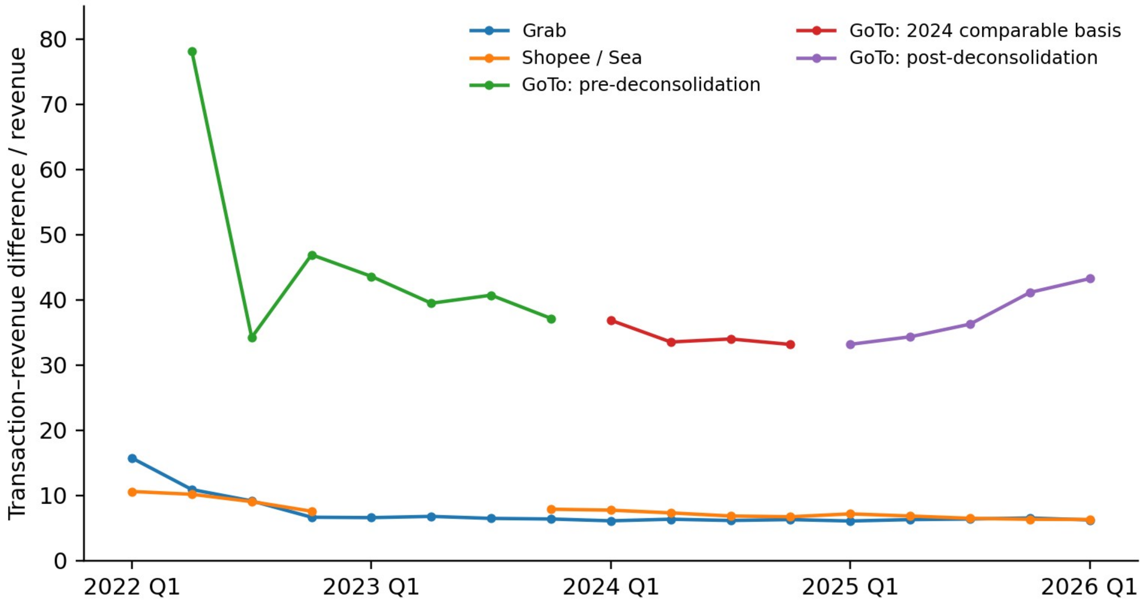

Figure 2. Company-wide quarterly accounting ratios

Notes: Ratios are calculated from the retained quarterly inputs. Missing Sea quarters are not connected, and GoTo is separated before, during, and after the 2024 comparable/pro-forma reporting basis. The lines do not represent Indonesian country panels or causal estimates. Source: 47-row accounting panel; see Table 9 and Appendix C.

Grab's mean quarterly ratio declines from 10.61 in 2022 to 6.56 in 2023 and remains around 6.23–6.32 in 2024–2025. The early change cannot be attributed solely to commercial monetization because Grab's filings also discuss the delivery-business recognition change. Sea's retained mean declines from 9.35 in 2022 to 6.71 in 2025, with the incomplete 2023 observation explicitly separated. GoTo's mean is 40.22 in 2023, 34.38 on the comparable/pro-forma 2024 basis, and 36.22 in 2025. The 2024 deconsolidation prevents interpreting that sequence as an unchanged business perimeter.

The source-reconciliation files map all 47 records to archived primary documents, but their precision is not uniform. The Grab audit records two US$1 million GTV differences, retained as flags rather than silently corrected. Sea's input amounts are often rounded release figures. One GoTo quarter is supported only at the source's rounded-display precision, and FY2024 uses explicit comparable/pro-forma values. These facts justify limited displayed precision and descriptive use. They do not justify treating 47 repeated company records as 47 independent Indonesian observations or pooling them into a causal regression.

7. Interpretation, limitations, and further empirical work

7.1 From an accounting difference to an information question

The paper documents a distinction between the size of transactions and the selected revenue retained in financial reporting. Whether the corresponding participant activity is visible to a tax authority is a separate variable that the public company data do not observe. A restaurant may report its receipts, a seller may be covered by another reporting system, and some transactions may already carry relevant taxes. Nothing in equation (1) distinguishes these cases.

OECD (2020) model rules describe collecting and reporting information about income earned by platform sellers offering accommodation, transport, and personal services. The European Union's 2021 DAC7 instrument provides a related platform-reporting framework. These are institutional examples of the information channel, not evidence that the Indonesian baseline residual is missing from official records. The discussion uses the cited instruments as precedents and does not claim a comprehensive description of Indonesian law as of 2026.

A direct test would require participant-level identifiers and transaction information linked to declarations, withholding, or audit outcomes, with consistent treatment of fees, refunds, costs, thresholds, and periods. The estimand would then concern reporting completeness or liability under specified rules. The present difference could help organize the measurement discussion, but it would not substitute for that linkage. Likewise, multiplying US$40.070 billion by a proposed tax rate would not produce a defensible revenue estimate.

7.2 What the evidence adds, and what it does not

The empirical contribution is the reconstruction of a specified set of cases and a demonstration of the consequences of measurement choices. Geography determines whether a published number belongs in the main comparison. Revenue definitions materially alter ratio levels even for the same platform and year. Assumed country rates impose some of the ratios by construction. Source timing matters when annual comparisons combine pre-acquisition, pro-forma, and reported quantities.

These results are useful precisely because they limit interpretations that a large headline ratio might otherwise invite. They do not establish that the arithmetic residual is a new measure of informality. The archive's size and reproducibility improve inspection and error detection, but neither increases the number of independent observations nor provides identification of a tax-reporting effect. The same distinction applies to the historical data: repetition documents the operating-accounting relationship, not the invisibility of that activity to government.

Cross-platform aggregation remains conditional. The source set cannot eliminate overlapping payment and marketplace activity or make financial-service transaction metrics equivalent to e-commerce orders. The main cases are therefore reported separately before the arithmetic sum. No extrapolation to all Indonesian platforms or to ASEAN is included among the current results.

7.3 Investor response: an extension, not a completed result

An earlier research question asked whether investors respond to a larger transaction–revenue difference. The accounting panel is relevant to that question because it records quantities available in earnings disclosures. However, the submitted market-data module states that event-session and concurrent-news classifications are unfinished. This manuscript therefore makes no finding—positive or null—about cumulative abnormal returns (CAR). It does not carry forward the prior thesis's reported null result as though the latest audit had certified it.

Completing the extension would require verifying the first trading session able to react to each release, retaining the contemporaneous accounting definition, reconciling stock and benchmark returns, and fixing treatment of concurrent guidance, acquisitions, repurchases, and restructuring before comparing specifications. Repeated observations from only three firms and structural accounting breaks would remain limitations even after these checks. Standard event-study methods provide a framework for the timing and return calculations, not an automatic solution to these identification problems (Brown and Warner, 1985; MacKinlay, 1997).

More fundamentally, a ratio level is not necessarily new information. A pricing analysis would need to explain what investors could have anticipated and why a change in the ratio represents informative news beyond revenue, transaction growth, profitability, and guidance released at the same time. The current public data do not establish an expectations benchmark. The investor analysis is accordingly a proposed additional test requiring design review, and remains outside the reported empirical findings.

7.4 Regional material and research scope

The regional files are retained as contextual evidence, not pooled observations. A common take rate applied to several national market-size figures mechanically produces similar ratios and cannot constitute an independent cross-country robustness test. Moreover, different countries' digital-economy totals and tax-collection series can refer to different activities, years, and liabilities. The earlier ASEAN calibration and Malaysia–Vietnam comparison are not used here to validate the Indonesia measurement.

A regional extension would need comparable country-platform transaction and revenue inputs, equivalent measurement periods, and a clear account of any allocation assumptions. A policy evaluation would require an outcome and comparison design appropriate to the specific intervention. Preserving the source material leaves those extensions possible without presenting their unfinished identification as a current finding.

8. Conclusion

This paper reconstructs transaction–revenue differences for three FY2023 Indonesia-focused platform cases. Grab is based on direct country revenue and a Group-rate GTV proxy; Tokopedia uses a directly reported Indonesia-aligned e-commerce segment pair; Shopee combines external market estimates with a company-wide service monetization rate. Under these constructions, the selected cases give US$43.233 billion of transaction value, US$3.162 billion of selected revenue, and a US$40.070 billion difference. The 12.67 aggregate ratio is conditional on those definitions and assumptions, not a national structural parameter.

The more substantive finding is that the measurement boundary matters. The ecosystem ratio is a transformation of the monetization rate, and two of the country ratios are therefore imposed by the construction. Tokopedia's ratio changes from 39.30 to 26.68 when its gross rather than net revenue is used. Grab and Shopee likewise yield different company-wide ratios when business or revenue scope changes. Earlier and quarterly disclosures document variation over time but do not independently validate the Indonesian allocation assumptions.

The paper supports careful reconstruction and reporting of platform activity, while leaving tax-reporting completeness and investor response as separate empirical questions. Public financial statements do not reveal how much of the residual is unreported, untaxed, or absent from GDP. A defensible application begins with the precise platform, period, geographic connection, operating definition, and revenue boundary, and reports what remains assumed. Those conditions are necessary before the size of a transaction–revenue difference can inform a broader economic or policy claim.

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Appendix A. Variable definitions and source locators

Table A1 defines the analytical variables without assuming that the underlying company measures are interchangeable. Table A2 identifies the six baseline sources. Published units are preserved in Table 2; the monetary output tables use U.S. dollars.

Table A1. Variable definitions

Variable	Definition and use	Source
GTV (V)	Generic transaction-value notation. Preserve the issuer's original GMV or GTV definition; not a common value-added measure.	G1, G2, T1, S1; M1 for country market estimates
Selected revenue (R)	Grab: all Indonesia-attributed revenue. Tokopedia: third-party net e-commerce segment revenue. Shopee: derived e-commerce service revenue.	G1; T1; S1 and M1
Transaction–revenue difference (W)	V − R, computed within the documented construction. Not verified participant payouts, income, unpaid tax, or missing GDP.	Author calculation
Ecosystem ratio (E)	(V − R)/R = 1/t − 1, for positive R. Not interpreted when the denominator is non-positive.	Author calculation
Monetization rate (t)	Selected revenue divided by transaction value. May incorporate more than a contractual commission.	Author calculation; stated Sea rate S1
Blended rate	Sum of selected revenue divided by sum of selected transaction value. Applies only to the three-case arithmetic display.	Author calculation
Market size and share (M, s)	External Indonesian e-commerce estimate and Shopee share. Their product supplies the baseline country GTV estimate.	M1
Exchange rate (f)	Annual period-average IDR per U.S. dollar. IDR million / f / 1,000 gives US$ billion.	F1
Quarterly ratio	E computed from each retained company-quarter pair, preserving original business and consolidation scope.	Quarterly accounting file and source-reconciliation ledgers

Notes: Definitions describe this paper's calculations. They do not confer validity on a country-allocation assumption or make unlike company definitions equivalent.

Grab defines its GMV as the total dollar value of transactions from its products and services, including applicable taxes, tips, tolls, surcharges and fees; it also includes sales through offline stores. Its original Group figure includes Financial Services. Sea defines GMV as the value of orders of products and services on the Shopee marketplace, including shipping and other charges; its order definition does not require settlement or exclude returned items. GoTo's operating table distinguishes e-commerce, on-demand services and financial technology and reports intercompany eliminations. This paper uses the e-commerce line only for Tokopedia. These definitions rule out treating the combined GTV display as a reconciled measure of final consumption or value added (Grab, 2024a; Sea, 2024a; GoTo, 2024).

Table A2. Baseline source documents

Key	Document / file	Locator and application
G1	Grab FY2023 Form 20-F grab_2023_20f.htm	Country revenue paragraph: Indonesia US$605m (archived HTML line 1002). Group revenue US$2,359m (line 1263). GMV definition and agent/principal discussion in the same filing.
G2	Grab FY2023 results release grab_2023_results.htm	Operating table: Group GMV US$20,983m (HTML lines 282–285). On-Demand GMV and Mobility/Deliveries segment tables supply the additional scope comparison.
T1	GoTo Annual Report 2023 goto_annual_report_2023.pdf	E-commerce GTV: printed p.142 / PDF p.144. Net revenue, gross revenue and incentives: note 29, financial p.5/111 / PDF p.475. Indonesia business description: printed p.72 / PDF p.74.
S1	Sea FY2023 Form 20-F sea_2023_20f.htm	Marketplace description: US$78.5bn GMV (HTML line 7612). Service revenue and stated 10% rate: lines 17584–17585. Revenue breakdown and segment-information tables supply precise service and total e-commerce revenue.
M1	Katadata Databoks, Santika (2025) katadata_momentumworks_indonesia_2023.htm	Embedded article data, archived HTML line 169: FY2023 Indonesia market US$53.8bn; Shopee share 40%. External estimates attributed to Momentum Works.
F1	World Bank WDI API response world_bank_idn_official_exchange_rate_2023.json	Country IDN; indicator PA.NUS.FCRF; date 2023; numeric value 15236.884662.

Notes: All six documents are retained in 01_RAW_SOURCES/main_indonesia. HTML line references describe the archived copy and may not match a reformatted website. PDF page numbers are one-based. The GTV and revenue locators in T1 are separate; the earlier input file's GTV locator is corrected here, without changing the underlying amount.

Appendix B. Historical coverage and exclusions

Table B1 records the nine platform-years in the Indonesia extension. Three are repetitions of the FY2023 main cases, not additional observations. One FY2021 Tokopedia record is withheld because the transaction and revenue periods do not match. The remaining rows retain their original derived or segment-level status; inclusion here does not validate their country allocation.

Table B1. Indonesia extension coverage

Year	Platform	E	Construction	Use in this draft
2021	Grab	22.79	Group-rate GTV proxy	Supporting only
2021	Tokopedia	Withheld	Full-year pro-forma GTV / post-acquisition revenue	Period mismatch
2022	Grab	12.91	Group-rate GTV proxy	Supporting only
2022	Shopee	10.90	External market / service rate	Supporting only
2022	Tokopedia	66.76	Segment GTV / net revenue	Supporting only
2023	Grab	7.89	Group-rate GTV proxy	Baseline overlap
2023	Shopee	9.00	External market / service rate	Baseline overlap
2023	Tokopedia	39.30	Segment GTV / net revenue	Baseline overlap
2024	Shopee	8.26	External market / service rate	Supporting only

Notes: Ratios are unit-free. The nine archived records do not form a balanced or fully harmonized country panel. The FY2021 Tokopedia ratio of 114.46 in the archive is not an active result in this manuscript. GoTo (2023), note 33, PDF p.494, identifies the revenue amount as a contribution after the 17 May 2021 acquisition. No replacement full-year revenue is inferred.

The broader archive also contains twenty Grab country-year revenue entries and forty-one historical source-extract rows. The former document available country revenue disclosure, not country GTV. The latter feed the eight annual records in Table 8 and are not forty-one independent annual ratios. The source and output counts serve different audit purposes and are never added to the main sample size.

The three FY2023 baseline cases, seven interpretable earlier company-wide ratios, and forty-seven quarterly accounting records remain distinct evidence sets. Their existence alone does not establish a tax-reporting or asset-pricing result. New country or period observations should enter only when both sides of the comparison can be described on a compatible basis.

Appendix C. Reproduction and unresolved empirical work

The source archive used for this draft is Invisible_Ledger_Empirical_Data_for_Professor_Kong_2026-09-09.zip (141,744,184 bytes). The raw source files, source-extract CSVs and original result files are preserved. Rebuilding a working copy with the supplied calculation script reproduced the archived result CSVs without changes. This confirms computational reproduction of those files, not validation of the economic assumptions.

Table C1. Data-to-result map

Input or retained file	Manuscript use
02_SOURCE_EXTRACTS/fy2023_indonesia_source_inputs.csv	Tables 2–3; baseline formulas in Section 4
05_RESULTS/fy2023_indonesia_main_one_way_sensitivity.csv	Table 4; fourteen scenario rows
05_RESULTS/fy2023_indonesia_leave_one_platform_out.csv	Table 5; three composition checks
05_RESULTS/historical_within_platform_ratios.csv	Table 8; eight records, seven usable ratios
05_RESULTS/indonesia_platform_year_extension.csv	Table B1; FY2021 Tokopedia result withheld
07_EVENT_AND_MARKET/ clean_event_panel_accounting.csv	Table 9 and Figure 2; no CAR result
09_QUARTERLY_SOURCE_AUDIT/ quarterly_panel_source_coverage.csv	Source mapping for all retained quarterly records
Additional extractions from T1, G2 and S1; official BPS GDP release	Tables 6–7 and Section 5.4; separately documented additions

Notes: The accompanying calculation notes record additional source inputs, calculations, and exclusions used in this draft. The main eleven-input file is not overwritten. Individual issuer-release URLs remain in the quarterly source-coverage CSV; the compiled company-year table does not replace that ledger.

The quarterly audit records complete document mappings but heterogeneous precision. Grab has fifteen exact GTV matches and two differences of US$1 million, with all seventeen revenue values reconciled. Sea's fourteen pairs match the archived release figures at their reported precision. GoTo has twelve exact, three rounding-confirmed and one rounded-display-confirmed records. Its 2024 values explicitly use a comparable/pro-forma basis. These classifications are properties of the retained reconciliation files, not a claim that company metrics have been economically harmonized.

The stock-price files and event ledgers are retained but not used for a completed investor test. Timing and concurrent-news classifications remain unresolved. Before reporting CAR, the return data, event-session rule, information set and contamination treatment would need a separate audit and an agreed design. No historical p-value or null conclusion is carried into this manuscript.

Advisor decisions on the main sample, aggregation and any additional empirical test remain pending. Reproducing the arithmetic does not establish the sufficiency of the final thesis.